LEARNING MODULE

23

Linear Regression

A regression coefficient is a measurement with an error bar. The error bar depends on what the errors do, and one significant intercept among 28 sectors is what chance delivers.

VOLUME IV · Statistical Inference and Econometrics

BUILDS ON LM2 · LM8 · LM10 · LM11 · LM12 · LM16 · LM17 · LM18 · LM19 · LM20 · LM21

LEADS TO LM25 · LM26 · LM27 · LM28 · LM30 · LM33 · LM34

TIME ≈ 3 h + lab

LEARNING OUTCOMES

Mastery	*After this module you should be able to:*
☐	**1** write the linear regression model in matrix form, compute OLS estimates, residuals and $s_\varepsilon^2 = \mathrm{RSS}/(T-d)$, and state the assumptions under which OLS is unbiased and best among linear unbiased estimators
☐	**2** compute standard errors, t-stats, confidence intervals and prediction intervals for a regression, identify days with large hat values, and interpret R-squared as a statistic
☐	**3** explain why heteroskedastic or autocorrelated errors invalidate the classical standard errors, and compute White and Newey–West standard errors, choosing the lag and checking its effect
☐	**4** estimate a market model for a sector index, interpret its beta and intercept, and explain why a raw-return intercept on price indices is not Jensen's alpha
☐	**5** size a hedge by regression, and judge a beta's stability across halves, calendar years and rolling windows
☐	**6** test 28 sector alphas with Bonferroni, Holm and Benjamini–Hochberg, and explain how correlation across sectors changes the count of false positives
☐	**7** estimate a factor model with a factor named in advance, and diagnose the collinearity it brings
☐	**8** compute OLS with classical and robust standard errors in NumPy, and check them against statsmodels with every option passed explicitly

LEARNING MODULE OVERVIEW

- OLS solves $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}$, (23.1). It is unbiased when the errors have mean zero given X, and its variance is $\sigma_\varepsilon^2 (X^\top X)^{-1}$ when they are also homoskedastic and uncorrelated, (23.2). The estimate $s_\varepsilon^2 = \mathrm{RSS}/(T-d)$ is unbiased, (23.3).
- Semiconductor's market model on 4,382 daily price returns has beta 1.207 and intercept 2.10 bp a day (5.16% a year). The beta's standard error is 0.0088 classical, 0.0129 White and 0.0185 Newey–West.
- A day far from the average return has a large hat value: 7 April 2025 had 21 times the average in FinancialInsurance's 2025 fit, and moved its beta by 2.8 classical standard errors.
- White's sandwich (23.6) allows heteroskedastic errors. On LM18's model paths the classical standard error is 43% of the true one.
- Overlapping 20-day sums make residuals autocorrelated (0.953 at lag 1). Newey–West (23.7) allows for it, and a standard error that keeps growing with the lag signals a drifting beta.
- A raw intercept on price indices mixes Jensen's alpha with interest and dividend terms, (23.8). The halves' betas differ by more than 1.96 Newey–West standard errors in 25 of 28 sectors, and FinancialInsurance's first-half hedge ratio fails in the second half.
- Of 28 sector intercepts one clears 1.96, against 1.4 expected by chance; Bonferroni, Holm and Benjamini–Hochberg reject none. Correlated sectors keep the expected count but change its spread.
- An electronics factor named in advance raises Semiconductor's R-squared from 0.8112 to 0.9163. The variance inflation factor (23.9) is 16.7 for the raw pair of regressors and 1.15 for the factor.

1 INTRODUCTION

LM12 fitted lines by least squares and read their coefficients as geometry. Here the same coefficients are estimates, which another sample would change. Each gets a standard error, which depends on what the model's errors do.

MOTIVATING CASE **Is Semiconductor's alpha real?**

From 1 December 2008 to 8 October 2026 the TWSE Semiconductor price index rose 38.1-fold, against 10.9-fold for the TAIEX price index. Regress Semiconductor's 4,382 daily returns on the TAIEX's. The slope, the sector's beta, is 1.207. The intercept is 2.10 bp a day, 5.16% a year, with a classical t-stat of 2.10. That clears the familiar 1.96 (Exhibit 1).

Before calling it skill, answer three questions. How sure is the beta? Its standard error is 0.0088 by the textbook formula, 0.0129 by White's and 0.0185 by Newey–West's. The beta was 1.089 in the sample's first half and 1.294 in the second (Newey–West standard errors 0.023 and 0.029). What does the intercept measure, when price indices omit dividends and the regression omits interest? And how many sectors were tried? Of 28, one intercept clears 1.96; chance alone predicts 1.4.

Exhibit 1: Semiconductor on the TAIEX: estimates and three standard errors

Statistic	Estimate	SE classical	SE White	SE Newey–West
Beta	1.207	0.0088	0.0129	0.0185
Intercept (bp a day)	2.10	1.00	1.00	1.01
t-stat of the intercept		2.10	2.10	2.09
Sectors of 28 with $\lvert\text{t-stat}\rvert > 1.96$		1	1	2

Note: U28's daily simple price returns (no dividends), $T = 4{,}382$; in-sample; Newey–West with $\ell = 9$. R-squared 0.8112. The 28 sectors' betas run from 0.507 (CommunicationsInternet) to 1.207 (Semiconductor). Source: FinMind `TaiwanStockPrice`, retrieved 11 October 2026, with 524 closes on 38 dates of U28 repaired from the 13:30:00 values of `TaiwanStockEvery5SecondsIndex` (`data/vol2_series.py`). Computations: `data/lm23_numbers.py`.

Sections 2–5 build the model and its classical and robust standard errors. Sections 6–8 apply them to betas, hedges, the 28 intercepts and a second factor; Section 9 uses Python.

NOTE **Notation for this module**

The data are U28: $T = 4{,}382$ daily simple price returns (a count) of $N = 28$ TWSE sector indices and the TAIEX price index, 2 December 2008 to 8 October 2026. Over 17.8508 years that gives $A = 245.4794$ sessions a year; each half holds 2,191 returns. Sector i, the TAIEX and the Electronic index return $R_{i,t}$, $R_{\text{mkt},t}$ and $R_{\text{el},t}$.

$\mathbf{y}$, X, $\boldsymbol{\beta}$, $\hat{\boldsymbol{\beta}}$, $\hat{\mathbf{y}}$, $\boldsymbol{\varepsilon}$, $\hat{\boldsymbol{\varepsilon}}$, Π: response; $T \times d$ design with first column $\mathbf{1}$ and row t equal to $\mathbf{x}_t^\top$ (LM12's $(\mathbf{1}\ X)$); coefficients β_j; estimates; fitted values; errors ε_t; residuals $\hat{\varepsilon}_t$; projection matrix (12.2). A market model's coefficients are α and β. Local: σ_ε^2, Σ_ε error variance and $\text{Var}(\boldsymbol{\varepsilon} \mid X)$; $\text{RSS} = \|\hat{\boldsymbol{\varepsilon}}\|^2$; s_x, $\bar{x}$, $\bar{y}$ a simple regression's regressor standard deviation and means; h_t, h_0 hat values; $\mathbf{x}_0$, y_0, ε_0 a new day's regressors, response and error; $\mathbf{a}$ a weight vector; $\beta_{j,\text{null}}$ a coefficient under H_0; Γ_k, $\hat{\Gamma}_k$ lag covariances of $\varepsilon_t \mathbf{x}_t$; ℓ the Newey–West lag; α_{J} Jensen's alpha; z the halves' standardized change; β_{mkt}, β_{el} two-factor coefficients (superscript raw on TAIEX and Electronic); q_i, q_{mkt} simple dividend yields a year; F_t TX's settlement price; C a correlation matrix; M, K generic matrices; counts: m days in a sum ($m = 20$; LM18's paths have $m = 1{,}231$) and k a lag. Descriptive s, s_x and correlations divide by T. An intercept a year is A times its daily value.

NOTE **Research hygiene**

Sector figures use U28's price returns, which leave out dividends, so every intercept includes a dividend-yield gap. Fits and tests are **in-sample**. On the same data the module tests 28 alphas (corrected in Section 7), 28 factor loadings, one beta against 1 and the 28 changes between the halves. On 0050 and 00631L it tests 12 slopes (Example 7).

Halves, years and rolling windows are **stability checks**. The first-half hedges are **out-of-sample**: each judged on the half its ratio did not see. LM18's paths and experiment 1 are **model** results. Correlations, cosines, R-squared values, variance cuts and q_{mkt} are **descriptive**, without standard errors: they describe this sample. The widget shows how far R-squared moves across halves and years.

2 THE LINEAR MODEL AND ITS ASSUMPTIONS

- [ ] write the linear regression model in matrix form, compute OLS estimates, residuals and $s_\varepsilon^2 = \text{RSS}/(T-d)$, and state the assumptions under which OLS is unbiased and best among linear unbiased estimators

LM12's four points, regressor values 0, 1, 2, 3 and responses $\mathbf{y} = (1, 3, 2, 6)^\top$, gave the line $\hat{\alpha} = 0.9$, $\hat{\beta} = 1.4$. Four other points from the same process would give another line. How far the line can move is a question about a model of the data.

Write the model and its least-squares estimate as

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad \hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}. \tag{23.1}$$

Here X is the $T \times d$ design with first column $\mathbf{1}$, and $\boldsymbol{\varepsilon}$ is the unobserved error. The estimate is LM12's solution of (12.1), the **ordinary least squares** (OLS) estimator. It is unique when X has full column rank, so that $X^\top X$ is invertible (LM10).

THEOREM **Mean and variance of OLS**

Let $\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, where

1. the design has full column rank d, and
2. the errors have conditional mean zero, $\mathbb{E}[\boldsymbol{\varepsilon} \mid X] = \mathbf{0}$.

Then $\mathbb{E}[\hat{\boldsymbol{\beta}} \mid X] = \boldsymbol{\beta}$. If also 3. the errors are homoskedastic and uncorrelated, $\operatorname{Var}(\boldsymbol{\varepsilon} \mid X) = \sigma_\varepsilon^2 I$,

then

$$\begin{aligned} \mathbb{E}[\hat{\boldsymbol{\beta}} \mid X] &= \boldsymbol{\beta}, \\ \operatorname{Var}(\hat{\boldsymbol{\beta}} \mid X) &= \sigma_\varepsilon^2 (X^\top X)^{-1}. \end{aligned} \tag{23.2}$$

Proof. Substituting the model gives $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (X^\top X)^{-1} X^\top \boldsymbol{\varepsilon}$, a matrix fixed given X times $\boldsymbol{\varepsilon}$. Its conditional expectation is $\mathbf{0}$ by condition 2, so $\mathbb{E}[\hat{\boldsymbol{\beta}} \mid X] = \boldsymbol{\beta}$. By (16.4) and condition 3,

$$\begin{aligned} \operatorname{Var}(\hat{\boldsymbol{\beta}} \mid X) &= (X^\top X)^{-1} X^\top (\sigma_\varepsilon^2 I) X (X^\top X)^{-1} \\ &= \sigma_\varepsilon^2 (X^\top X)^{-1}. \qquad \blacksquare \end{aligned}$$

The model itself is the linearity assumption; normal errors are an optional fourth. In a simple regression, $d = 2$, and the slope's entry of $(X^\top X)^{-1}$ is $1/(T s_x^2)$, so $\operatorname{Var}(\hat{\beta} \mid X) = \sigma_\varepsilon^2 / (T s_x^2)$. The variance falls like $1/T$, the rate LM18 assumed for a slope's batch errors.

The error variance is unknown. Its estimate divides the residual sum of squares by $T - d$:

$$s_\varepsilon^2 := \frac{\mathrm{RSS}}{T - d}, \qquad \mathbb{E}[s_\varepsilon^2 \mid X] = \sigma_\varepsilon^2. \tag{23.3}$$

The residual is $\hat{\boldsymbol{\varepsilon}} = (I - \Pi)\mathbf{y} = (I - \Pi)\boldsymbol{\varepsilon}$, because $\Pi X = X$. Since $I - \Pi$ is symmetric and idempotent, $\mathrm{RSS} = \boldsymbol{\varepsilon}^\top (I - \Pi) \boldsymbol{\varepsilon}$. Its expectation sums each entry of $I - \Pi$ times the matching entry of $\operatorname{Var}(\boldsymbol{\varepsilon} \mid X)$. By condition 3 those entries are σ_ε^2 on the diagonal and 0 off it, which gives $\sigma_\varepsilon^2 \operatorname{tr}(I - \Pi) = \sigma_\varepsilon^2 (T - d)$ by (12.2). For normal errors the maximum-likelihood estimate divides RSS by T, as LM20's normal MLE (20.5) does.

EXAMPLE 1

Four points with an error bar

For LM12's four points ($\hat{\alpha} = 0.9$, $\hat{\beta} = 1.4$, $\mathrm{RSS} = 4.2$), compute s_ε^2, the standard errors of both coefficients and a 95% interval for β under normal errors.

SOLUTION

Here $T - d = 2$, so $s_\varepsilon^2 = 4.2/2 = 2.1$. The regressor has $\bar{x} = 1.5$ and $T s_x^2 = 5$. By (23.2), $\mathrm{SE}(\hat{\beta}) = \sqrt{2.1/5} = 0.648$ and $\mathrm{SE}(\hat{\alpha}) = \sqrt{2.1\,(1/4 + 1.5^2/5)} = 1.212$. The t-stat of $\hat{\beta}$ is $1.4/0.648 = 2.16$.

With normal errors the t-stat follows t_2 (Section 3), whose 97.5% quantile is 4.303. The interval $1.4 \pm 4.303 \times 0.648$ runs from −1.39 to 4.19 and contains 0. The normal quantile 1.96 would give 0.13 to 2.67 and exclude 0. With two residual degrees of freedom, the normal reference misleads.

THEOREM **Gauss–Markov**

Under the three conditions of (23.2), let $M\mathbf{y}$ be any linear estimator of $\boldsymbol{\beta}$ that is unbiased for every $\boldsymbol{\beta}$, with M a $d \times T$ matrix fixed given X. Then $\mathrm{Var}(M\mathbf{y} \mid X) - \mathrm{Var}(\hat{\boldsymbol{\beta}} \mid X)$ is positive semi-definite: OLS is the **best linear unbiased estimator** (Greene 2018, ch. 4; proof in the Deep Dive).

No linear unbiased estimator gives any combination $\mathbf{a}^\top \boldsymbol{\beta}$ a smaller variance. The theorem says nothing about nonlinear estimators, nor about errors that break condition 3, as daily returns do (Section 4).

DEEP DIVE **Why OLS is best among linear unbiased estimators** optional · click to expand

Since $\mathbb{E}[M\mathbf{y} \mid X] = MX\boldsymbol{\beta}$, unbiasedness for every $\boldsymbol{\beta}$ requires $MX = I$. Write $M = (X^\top X)^{-1} X^\top + K$; then $KX = 0$. By (16.4),

$$\begin{aligned} \mathrm{Var}(M\mathbf{y} \mid X) &= \sigma_\varepsilon^2 M M^\top \\ &= \sigma_\varepsilon^2 (X^\top X)^{-1} + \sigma_\varepsilon^2 K K^\top, \end{aligned}$$

because the cross terms $(X^\top X)^{-1} X^\top K^\top$ vanish with $KX = 0$. The extra term is positive semi-definite, since for every $\mathbf{a}$

$$\mathbf{a}^\top K K^\top \mathbf{a} = \| K^\top \mathbf{a} \|^2 \geq 0. \qquad \blacksquare$$

PITFALL **Regress returns, not prices**

Regressing one trending price series on another can show a strong fit with no relation at all. LM25 treats this spurious regression; this module regresses returns.

導讀 **迴歸是帶誤差的量測**

迴歸（regression）把資料寫成 $\mathbf{y} = X\beta + \varepsilon$，其中係數（coefficient）是要估計的常數，誤差（error）是看不到的隨機部分。普通最小平方法（ordinary least squares）的估計量（estimator）就是 LM12 用投影（projection）求得的解；只要誤差在給定 X 時平均為零，它就是不偏估計量（unbiased estimator）。殘差（residual）只剩 $T - d$ 個維度（dimension），所以誤差的變異數（variance）要用 $\mathrm{RSS}/(T-d)$ 估計。誤差若變異數齊一（homoskedastic）且互不相關（uncorrelated），高斯-馬可夫定理（Gauss–Markov theorem）保證 OLS 是最佳線性不偏估計量（best linear unbiased estimator）。這個保證只比較線性不偏的估計量，而且誤差變異數一旦逐日不同就不再成立。

KNOWLEDGE CHECK 1

LM12 fitted TAIEX's 22 daily returns of December 2008 on the 28 sectors with an intercept: 29 columns of rank 22, an exact fit. What is s_ε?

ANSWER

The estimate s_ε does not exist. The fit is exact, so $\mathrm{RSS} = 0$, and the rank leaves $22 - 22 = 0$ residual degrees of freedom, so (23.3) reads $0/0$. An exact fit has no error estimate, hence no standard errors: it is no evidence.

3 STANDARD ERRORS, INTERVALS AND HAT VALUES

- [ ] compute standard errors, t-stats, confidence intervals and prediction intervals for a regression, identify days with large hat values, and interpret R-squared as a statistic

Replace σ_ε^2 by s_ε^2 in (23.2): the square roots of the diagonal of $s_\varepsilon^2 (X^\top X)^{-1}$ are the **classical standard errors**. A coefficient divided by its standard error is a t-stat, as in LM21's test of a mean (21.1).

RESULT **t-stats and intervals**

Under the conditions of (23.2), test $H_0 : \beta_j = \beta_{j,\text{null}}$ with

$$\begin{aligned} \text{t-stat}(\hat{\beta}_j) &= \frac{\hat{\beta}_j - \beta_{j,\text{null}}}{\text{SE}(\hat{\beta}_j)}, \\ \text{interval} \;\; \hat{\beta}_j &\pm 1.96\,\text{SE}(\hat{\beta}_j). \end{aligned} \tag{23.4}$$

With normal errors the t-stat follows t_{T-d} under H_0 (stated without proof; it is the regression form of LM20's t_{T-1} law). For large T, with finite fourth moments, it is approximately $\mathcal{N}(0,1)$ without normality (stated without proof; (17.5) is the reason). The interval then covers β_j with probability close to 95%.

Later t-stats and p-values are two-sided with the normal reference. Daily returns may lack a finite fourth moment (LM19), so these normal references are approximations. Every later t-stat comes from a fit with $T - d \geq 183$ (the widget's 2026 window). The 97.5% quantile of t_{183}, 1.973, exceeds the normal 1.960 by 0.7%. Each t-stat names its standard error: classical, White or Newey–West.

EXAMPLE 2

The case by hand

Compute Semiconductor's classical standard error of beta from s_ε and s_x, the 95% intervals with the classical and Newey–West errors, and the intercept's p-value.

SOLUTION

The residual standard deviation is $s_\varepsilon = 0.66239\%$ with divisor $T - 2$ (0.66224% with divisor T), and the TAIEX's returns have $s_x = 1.137\%$. So

$$\begin{aligned} \text{SE}(\hat{\beta}) &= \frac{s_\varepsilon}{\sqrt{T}\, s_x} \\ &= \frac{0.66239\%}{\sqrt{4382} \times 1.137\%} = 0.0088. \end{aligned}$$

The classical interval $1.2073 \pm 1.96 \times 0.0088$ runs from 1.190 to 1.225. With the Newey–West error 0.0185 it runs from 1.171 to 1.244 (Exhibit 5).

The intercept's classical t-stat is 2.10, with p-value $2\Phi(-2.10) = 0.036$ by (21.2). It is one test of 28 on the same data, which Section 7 counts.

R-squared, (12.4), is 0.8112 here. As a statistic it has its own sampling error, and it never falls when a column is added. **Adjusted R-squared**, $1 - (1 - \text{R-squared})(T-1)/(T-d)$, charges for columns: 0.81119 against 0.81124 (from unrounded values). The **F statistic** tests that all coefficients but the

intercept are zero; with one regressor it is the squared t-stat, $137.20^2 = 18{,}824$. R-squared describes a fit and tests nothing: LM12's 0.99912 for TAIEX on its own sectors was an accounting identity.

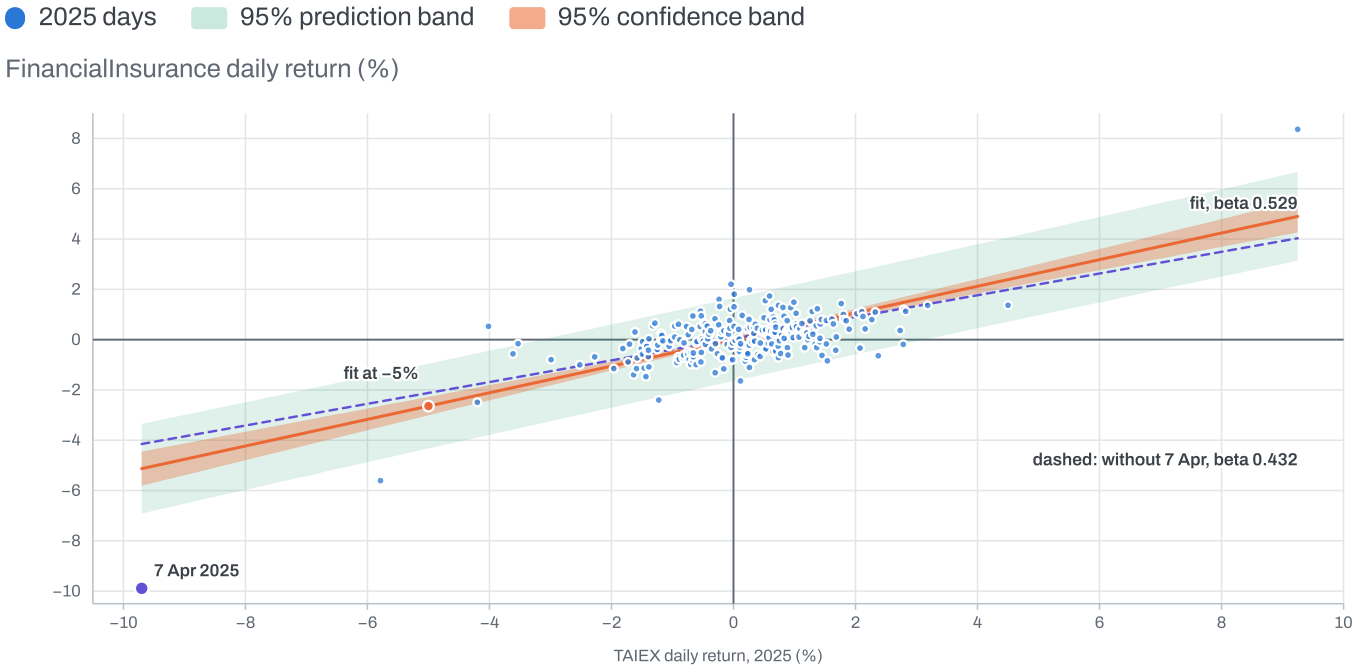
A new observation $y_0 = \mathbf{x}_0^\top \boldsymbol{\beta} + \varepsilon_0$ has an error of variance σ_ε^2, independent of the sample. By (16.4), the fitted value $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$ and the prediction error have

$$\begin{aligned} \operatorname{Var}(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} \mid X) &= \sigma_\varepsilon^2 h_0, \\ h_0 &:= \mathbf{x}_0^\top (X^\top X)^{-1} \mathbf{x}_0, \\ \operatorname{Var}(y_0 - \mathbf{x}_0^\top \hat{\boldsymbol{\beta}} \mid X) &= \sigma_\varepsilon^2 (1 + h_0). \end{aligned} \tag{23.5}$$

Under (23.2)'s conditions, with s_ε in place of σ_ε, ±1.96 standard errors give a **confidence interval** for the fitted line, approximately for large T. The **prediction interval** for a new day also needs normal errors, because one day's error does not average out.

At a sample point, $h_t := \mathbf{x}_t^\top (X^\top X)^{-1} \mathbf{x}_t$ is the diagonal entry of Π in row t, the **hat value** of date t. It is the weight of y_t in its own fitted value $\hat{y}_t$, a measure of influence. The hat values sum to $\operatorname{tr} \Pi = d$, so their average is d/T. Leverage keeps the book's meaning, $f > 1$.

Exhibit 2: FinancialInsurance on the TAIEX in 2025, with confidence and prediction bands



Note: 2025's 243 daily price returns, % a day; in-sample. Solid: the fitted line, $\hat{\beta} = 0.529$. Dashed: the fit without 7 April 2025, $\hat{\beta} = 0.432$. Bands: 95% confidence and prediction intervals by (23.5). On the full sample the confidence half-width at −5% is 0.105 points, too narrow to show at this scale. Source: as Exhibit 1.

EXAMPLE 3

One day in 2025

Fit FinancialInsurance on the TAIEX over the 243 returns of 2025 (Exhibit 2). Find the 95% confidence and prediction intervals at a TAIEX return of −5%, and the hat value and influence of 7 April 2025.

SOLUTION

The fit gives $\hat{\beta} = 0.529$, with classical standard error 0.0350, and $s_\varepsilon = 0.840\%$. At $\mathbf{x}_0 = (1, -0.05)^\top$, a fall that only 2 days of 2025 reached, the fitted return is −2.642% with standard error 0.1865 points. The confidence interval runs from −3.01% to −2.28%. The prediction standard error is 0.860 points, so the prediction interval, −4.33% to −0.96%, is over four times as wide.

On 7 April 2025 the TAIEX fell 9.70% and FinancialInsurance 9.89%. That day's hat value is $h_t = 0.1709$, against the average $d/T = 2/243 = 0.0082$: 21 times. Without it, $\hat{\beta} = 0.432$, so one day moves the year's beta by 2.8 classical standard errors. 0050 closed and TX settled at their −10% limits that day (LM2), so the limits may have capped the recorded moves.

導讀 **標準誤、信賴區間與 hat value**

係數（coefficient）除以它的標準誤（standard error）就是 t 統計量（t-statistic）；誤差（error）為常態分布（normal distribution）時它服從 t_{T-d}，樣本大時近似標準常態分布。估計值（estimate）加減 1.96 個標準誤，就是 95% 信賴區間（confidence interval）。判定係數（R-squared）只描述配適（fit），本身也有抽樣誤差（sampling error），不是檢定（test）。預測新的一天時，除了係數的誤差，還要加上當天自己的誤差，所以預測區間（prediction interval）寬得多。hat value 衡量一個點對自己配適值（fitted value）的影響力（influence）：2025 年 4 月 7 日的 hat value 是當年平均的 21 倍，這一天就讓全年的 beta 移動 2.8 個標準誤。

KNOWLEDGE CHECK 2

Test $H_0 : \beta = 1$ for Semiconductor ($\hat{\beta} = 1.2073$) with the classical standard error 0.0088 and with the Newey–West one, 0.0185. Do both reject at the 5% level?

ANSWER

By (23.4) the t-stats are $(1.2073 - 1)/0.0088 = 23.6$ and $(1.2073 - 1)/0.0185 = 11.2$. Both exceed 1.96 by far: Semiconductor moves more than one for one with the TAIEX, whichever standard error one trusts.

4 HETEROSKEDASTICITY AND WHITE'S STANDARD ERRORS

- ☐ explain why heteroskedastic or autocorrelated errors invalidate the classical standard errors, and compute White and Newey–West standard errors,

> choosing the lag and checking its effect

The classical standard errors rest on condition 3 of (23.2): every day's error has the same variance. Daily returns break it: large errors come on days of large index moves, far from the regressor's mean, where they move the slope most.

Keep conditions 1 and 2, and let $\Sigma_\varepsilon := \operatorname{Var}(\boldsymbol{\varepsilon} \mid X)$ be any covariance matrix. Applying (16.4) to $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (X^\top X)^{-1} X^\top \boldsymbol{\varepsilon}$ gives the **sandwich**

$$\operatorname{Var}(\hat{\boldsymbol{\beta}} \mid X) = (X^\top X)^{-1} X^\top \Sigma_\varepsilon X (X^\top X)^{-1}. \tag{23.6}$$

The outer factors are the bread and the middle the filling. When the errors are uncorrelated, Σ_ε is diagonal and the filling is $\sum_t \operatorname{Var}(\varepsilon_t \mid X)\, \mathbf{x}_t \mathbf{x}_t^\top$.

White's estimator replaces each unknown variance by the squared residual, so the filling becomes $\sum_t \hat{\varepsilon}_t^2 \mathbf{x}_t \mathbf{x}_t^\top$ (statsmodels' HC0). One squared residual estimates its variance badly; the sum over T days averages the errors out. White (1980) shows that the estimate is consistent for independent rows with finite moments as T grows (stated without proof). Variance that changes from day to day is what LM26 models; White's standard errors need no model of it.

EXAMPLE 4

LM18's slope on 10,000 paths

LM18 simulated 10,000 five-year paths of iid normal log returns ($m = 1{,}231$, `default_rng([3, 20261011])`) and regressed the growth factor G on $\ln G$. It reported $\hat{\beta} = 2.18$ with batch standard error 0.01. Compare the classical, White and batch standard errors with the model's value.

SOLUTION

OLS gives $\hat{\beta} = 2.1759$, with classical standard error 0.0067 and White's 0.0152. LM18's batch value from 20 batches of 500 is 0.0113. In the model $\ln G$ is exactly normal, so (23.6) with the model's moments in place of the sample's, for 10,000 paths, is an integral over a normal law. Quadrature gives 0.0157 around the true slope $\mathbb{E}G = 2.1584$.

The classical formula gives 43% of that value, and White's 0.97 of it. The batch value is 0.72 of it. By LM18's rule, a standard deviation of 20 values errs by about $1/\sqrt{40} = 16\%$, so the batch value misses by under two such errors.

Since $G = e^{\ln G}$ is convex, the line misses most at both ends of $\ln G$, where the regressor is farthest from its mean. The residuals' root mean square is 0.260, 0.179 and 0.378 in the lower, middle and upper thirds of $\ln G$ (cut at $\hat{F}^{-1}(1/3)$ and $\hat{F}^{-1}(2/3)$). The classical formula ignores where the large errors sit.

Across U28's 28 market models, White's standard error of beta is 1.19 to 1.91 times the classical one. In FinancialInsurance's 2025 fit (Example 3) it is 0.106, three times the classical 0.0350, and 0.097 without 7 April 2025. That year's errors are largest on its few days of large index moves.

So Example 3's bands are too narrow at −5%: White's standard error of the fitted value there is 0.557 points, not 0.1865. The two 2025 days with the TAIEX at or below −5% have residuals of −5.7 and −3.0 s_ε, outside the prediction band.

KNOWLEDGE CHECK 3

LM16's slope of FinancialInsurance on Semiconductor, 0.4591, has standard errors 0.0099 classical, 0.0165 White and 0.0229 Newey–West ($\ell = 9$). Give the Newey–West 95% interval. LM12 reported a residual standard deviation of 0.996%. How does it relate to s_ε?

ANSWER

The interval is $0.4591 \pm 1.96 \times 0.0229$, from 0.414 to 0.504. LM12 divided by T and got 0.99603%; (23.3) divides by $T - 2$ and gives 0.99626%. They differ by the factor $\sqrt{4382/4380}$, and both round to 0.996%.

5 OVERLAPPING RETURNS AND NEWEY–WEST

- [ ] explain why heteroskedastic or autocorrelated errors invalidate the classical standard errors, and compute White and Newey–West standard errors, choosing the lag and checking its effect

A regression of 20-day returns can start a sum on every day: days 1–20, 2–21 and so on. Neighbouring sums share 19 days, so their errors are autocorrelated, and condition 3 fails again. Semiconductor's 4,363 overlapping log-return windows leave residuals with sample autocorrelation (19.5) $\hat\rho_1 = 0.953$, close to the 19/20 of iid days.

With autocorrelated rows the filling of (23.6) also collects covariances between days. Divided by T, it tends to the **long-run variance** of the products $\varepsilon_t \mathbf{x}_t$,

$$\Gamma_0 + \sum_{k \ge 1} \left(\Gamma_k + \Gamma_k^\top \right), \qquad \Gamma_k := \mathbb{E}\left[\varepsilon_t \varepsilon_{t-k} \mathbf{x}_t \mathbf{x}_{t-k}^\top \right],$$

the matrix form of (19.7)'s limit. Newey and West (1987) estimate it with declining weights:

$$\hat\Gamma_0 + \sum_{k=1}^{\ell} \left(1 - \frac{k}{\ell + 1} \right) \left(\hat\Gamma_k + \hat\Gamma_k^\top \right), \qquad \hat\Gamma_k := \frac{1}{T} \sum_{t=k+1}^{T} \hat\varepsilon_t \hat\varepsilon_{t-k} \mathbf{x}_t \mathbf{x}_{t-k}^\top, \tag{23.7}$$

which replaces $X^\top \Sigma_\varepsilon X / T$ in (23.6). statsmodels calls $1 - k/(\ell + 1)$ Bartlett weights. With them the estimate is positive semi-definite. It is consistent when the products' dependence fades fast enough,

their moments are bounded and ℓ grows more slowly than $T^{1/4}$ (both stated without proof; Newey and West 1987).

The book fixes the lag in advance as $\ell = \lfloor 4(T/100)^{2/9} \rfloor$, whose exponent 2/9 is below 1/4: 9 for 4,382 returns, 7 for a half, 4 for a year. It is statsmodels' default HAC lag and the pilot lag of Newey and West (1994), not the data-based lag they recommend. A fixed rule avoids choosing the lag that gives the wanted answer.

EXAMPLE 5

Overlapping 20-day sums

Regress Semiconductor's overlapping 20-day log-return sums on the TAIEX's (4,363 windows), and compare the standard errors in Exhibit 3.

SOLUTION

The slope is 1.157 (daily: 1.204). Its classical standard error, 0.0089, is about the daily one, 0.0088, as if the overlapping sums were independent. Newey–West with $\ell = 19$, the overlap $m - 1$, gives 0.0389, about 4.4 times the classical value; $\ell = 38$ gives 0.0464.

The 219 non-overlapping sums from the first return give 1.106, with classical standard error 0.0406. Over the 20 possible starting days that slope runs from 1.106 to 1.189, about twice its standard error.

Under iid days the true factor would be 3.65, and Newey–West with $\ell = 19$ would show about 3.17 (Deep Dive). Heteroskedastic daily errors and a drifting beta (Exhibit 4) can explain why Semiconductor's 4.4 is larger.

Exhibit 3: Three sectors on 20-day sums: one slope, five standard errors

Log returns, 20-day sums	Semiconductor	FinancialInsurance	Food
Daily beta (classical SE)	1.204 (0.0088)	0.824 (0.0103)	0.538 (0.0117)
Overlapping beta, 4,363 windows	1.157	0.801	0.444
SE classical	0.0089	0.0106	0.0109
SE White	0.0115	0.0164	0.0159
SE Newey–West, $\ell = 19$	0.0389	0.0543	0.0535
SE Newey–West, $\ell = 38$	0.0464	0.0610	0.0628
Non-overlapping beta, 219 sums (classical SE)	1.106 (0.0406)	0.867 (0.0509)	0.518 (0.0515)
Non-overlapping beta, over the 20 starting days	1.106 to 1.189	0.707 to 0.889	0.387 to 0.518

Note: U28's log price returns, summed over 20 days; in-sample, the starting days a stability check. Source: as Exhibit 1.

DEEP DIVE **Why overlapping sums need a long lag** optional · click to expand

Let the daily errors and the regressor's daily returns be iid with mean zero and variances σ_ε^2 and σ_x^2, the two series independent. Two sums of m days that end $k < m$ days apart share $m - k$ days. By (16.4) the error sums then have covariance $(m-k)\sigma_\varepsilon^2$, and the regressor sums $(m-k)\sigma_x^2$. By (16.1) the slope's entry of Γ_k is their product:

$$\begin{matrix}(m-k)^2\sigma_\varepsilon^2\sigma_x^2 & (k<m),\\ 0 & (k\geq m).\end{matrix}$$

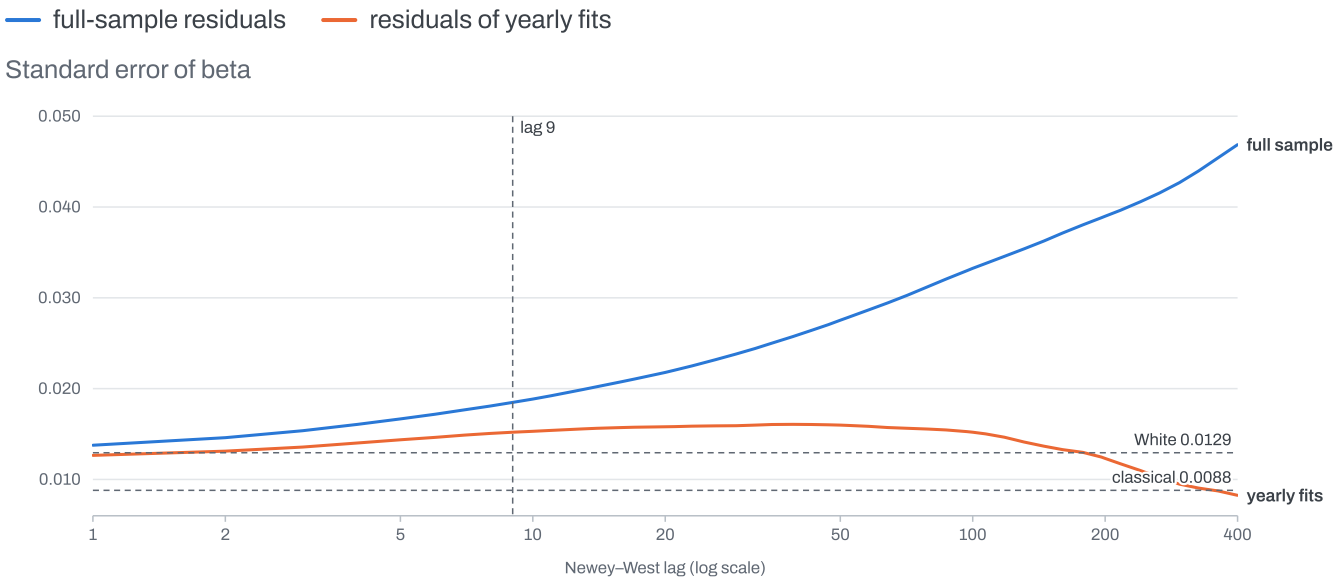
The classical formula keeps Γ_0 alone, $m^2\sigma_\varepsilon^2\sigma_x^2$. The long-run variance adds the other lags,

$$\begin{aligned}&2\sum_{k=1}^{m-1}(m-k)^2\sigma_\varepsilon^2\sigma_x^2\\ &\quad=\frac{(m-1)m(2m-1)}{3}\,\sigma_\varepsilon^2\sigma_x^2,\end{aligned}$$

so the true variance is $(2m^2+1)/(3m)$ times the classical one. At $m = 20$ that is 13.35, a factor 3.65 in the standard error. A lag below $m - 1$ drops terms, and Bartlett weights shrink the rest: in large samples $\ell = 19$ recovers a factor 3.17, and $\ell = 38$ a factor 3.41.

The lag also diagnoses a drifting beta (Exhibit 4). With the full-sample residuals, Semiconductor's daily Newey–West standard error keeps growing: 0.0129 at $\ell = 0$ (White's), 0.0185 at 9, 0.0275 at 50 and 0.0390 at 200. With a drifting beta, the products $\hat{\varepsilon}_t\mathbf{x}_t$ of the full-sample fit keep their sign for long stretches. Residuals from each calendar year's own line give 0.0152, 0.0160 and 0.0123 at lags 9, 50 and 200 instead.

Exhibit 4: Newey–West standard error of Semiconductor's beta against the lag



Note: U28's daily price returns, with the full-sample bread $(X^\top X)^{-1}$ in (23.6). The second series uses the residuals of one fit per calendar year (19 fits; 2008's 22 returns form their own), a diagnostic rather than an estimator. Their products sum to zero within each year, which pulls that series down at lags near a year. Dashed: the classical 0.0088, White's 0.0129 and the book's $\ell = 9$. Source: as Exhibit 1.

PITFALL **A standard error that grows with the lag**

When a Newey–West standard error keeps growing with ℓ, the constant coefficient itself is in doubt: the full-sample beta averages a moving target (Section 6). On overlapping sums, use $\ell \geq m - 1$ or non-overlapping sums, and report both.

LM22's block bootstrap is the resampling alternative to (23.7), and LM33 applies the long-run variance to a Sharpe ratio.

導讀 **變異數非齊一與自我相關**

古典標準誤（classical standard error）假設每天誤差（error）的變異數（variance）相同、互不相關（uncorrelated）。日報酬違反前一項：大盤大漲大跌的日子誤差也大，這就是變異數非齊一性（heteroskedasticity）。White 的夾層估計量（sandwich estimator）用殘差（residual）平方取代未知的變異數，得到穩健標準誤（robust standard error）。重疊的 20 日報酬相鄰兩個視窗（window）共用 19 天，殘差有很強的自我相關（autocorrelation）；Newey–West 的變異數非齊一與自我相關一致估計量（HAC）再把各落期（lag）的共變異數（covariance）加權納入。落期越長、標準誤一路上升，表示 beta 本身有參數漂移（parameter drift），問題不只出在標準誤。

KNOWLEDGE CHECK 4

Under iid days, what is the correlation of two overlapping 20-day sums that end k days apart? Evaluate it at k = 1, 10 and 20, and compare Semiconductor's overlap residuals.

ANSWER

The sums share $20-k$ days, so by (16.4) their covariance is $(20-k)\sigma_\varepsilon^2$ and their correlation $(20-k)/20$: 0.95, 0.50 and 0. Semiconductor's residuals show $\hat{\rho}_k$ of 0.953, 0.538 and 0.038 at lags 1, 10 and 20.

6 THE MARKET MODEL: BETA, ALPHA AND HEDGING

- ☐ estimate a market model for a sector index, interpret its beta and intercept, and explain why a raw-return intercept on price indices is not Jensen's alpha
- ☐ size a hedge by regression, and judge a beta's stability across halves, calendar years and rolling windows

The **market model** regresses a sector's return on the market's:

$$R_{i,t} = \alpha + \beta R_{\mathrm{mkt},t} + \varepsilon_t .$$

The market model's slope estimates the sector's beta, the population slope of (16.8). The model describes co-movement and makes no claim about expected returns.

The capital asset pricing model (CAPM) makes one. With total returns and a risk-free rate r_f, it states $\mathbb{E}[R_i] - r_f = \beta_i(\mathbb{E}[R_{\mathrm{mkt}}] - r_f)$ for every asset (Campbell, Lo and MacKinlay 1997, ch. 5; Sharpe 1964 and Lintner 1965 are its origins). **Jensen's alpha** α_{J} is the market model's intercept on total excess returns (Jensen 1968), zero under the CAPM.

A total return adds the dividend yield, q_i/A a day if spread evenly, so Jensen's regression reads

$$R_{i,t} + \frac{q_i - r_f}{A} = \alpha_{\mathrm{J}} + \beta\left(R_{\mathrm{mkt},t} + \frac{q_{\mathrm{mkt}} - r_f}{A}\right) + \varepsilon_t .$$

Collecting constants gives U28's raw intercept

$$\alpha = \alpha_{\mathrm{J}} + \frac{(1-\beta)\, r_f}{A} - \frac{q_i - \beta q_{\mathrm{mkt}}}{A} \tag{23.8}$$

per day, with r_f, q_i and q_{mkt} simple rates a year.

Over U28's window the TAIEX total-return index rose 20.23-fold and the price index 10.91-fold. Their mean daily return gap, times A, gives $q_{\mathrm{mkt}} = 3.46\%$ a year. For Semiconductor ($\beta = 1.207$), a 2% risk-free

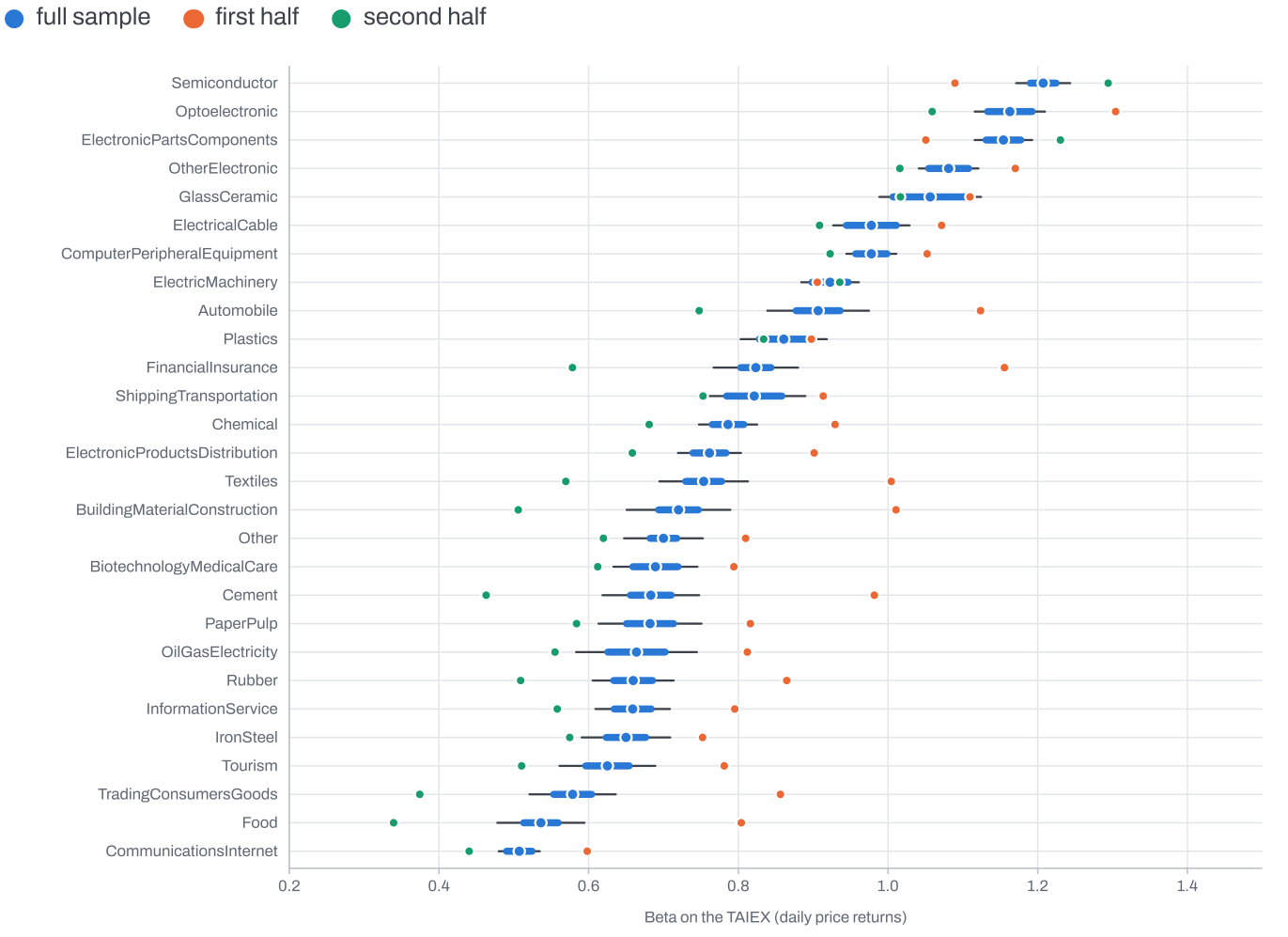
rate, LM6's proxy, adds 0.17 bp a day to Jensen's alpha, and equal dividend yields subtract 0.29 bp. Jensen's alpha is then 1.98 bp. Each point of sector yield below the index's subtracts another 0.41 bp; the sectors' own yields are not in the data.

A beta is an average over the sample. Exhibit 5 sets the 28 full-sample betas beside the halves'. With Newey–West errors ($\ell = 7$ in each half), the change between the halves in standard errors,

$$z = \frac{\hat{\beta}_{\text{second}} - \hat{\beta}_{\text{first}}}{\sqrt{\text{SE}_{\text{first}}^2 + \text{SE}_{\text{second}}^2}},$$

exceeds 1.96 in absolute value for 25 of 28 sectors. FinancialInsurance fell from 1.156 to 0.578 and Cement from 0.982 to 0.463; Semiconductor rose from 1.089 to 1.294. The halves' betas correlate 0.653.

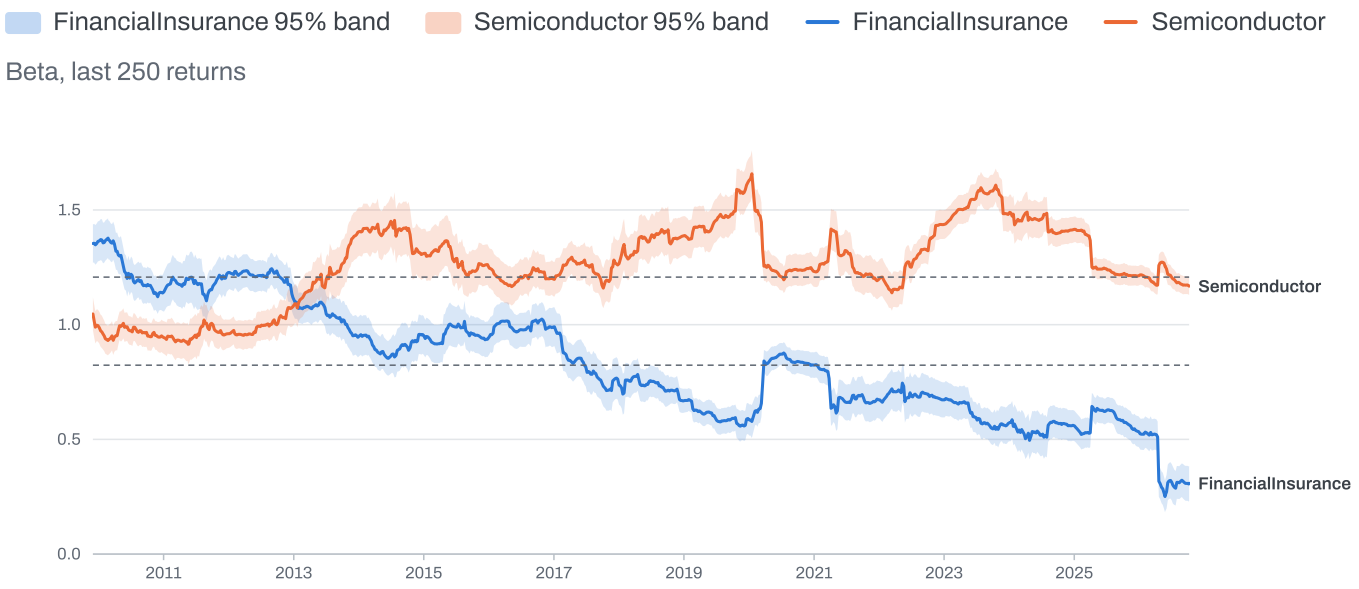
Exhibit 5: The 28 betas, full sample and halves



Note: market models of U28's daily price returns, sorted by the full-sample beta. Thick lines: classical 95% intervals of the full-sample beta (Semiconductor 1.190 to 1.225). Thin grey lines: Newey–West, $\ell = 9$ (1.171 to 1.244). The halves (2,191 returns each) are a stability check. Source: as Exhibit 1.

Exhibit 6 follows two betas over rolling 250-return windows. FinancialInsurance's fell from 1.377 (window ending 23 February 2010) to 0.251 (26 May 2026), and ends at 0.308. Semiconductor's ranged from 0.914 (18 May 2011) to 1.658 (17 January 2020), and ends at 1.168.

Exhibit 6: Rolling 250-return betas of FinancialInsurance and Semiconductor

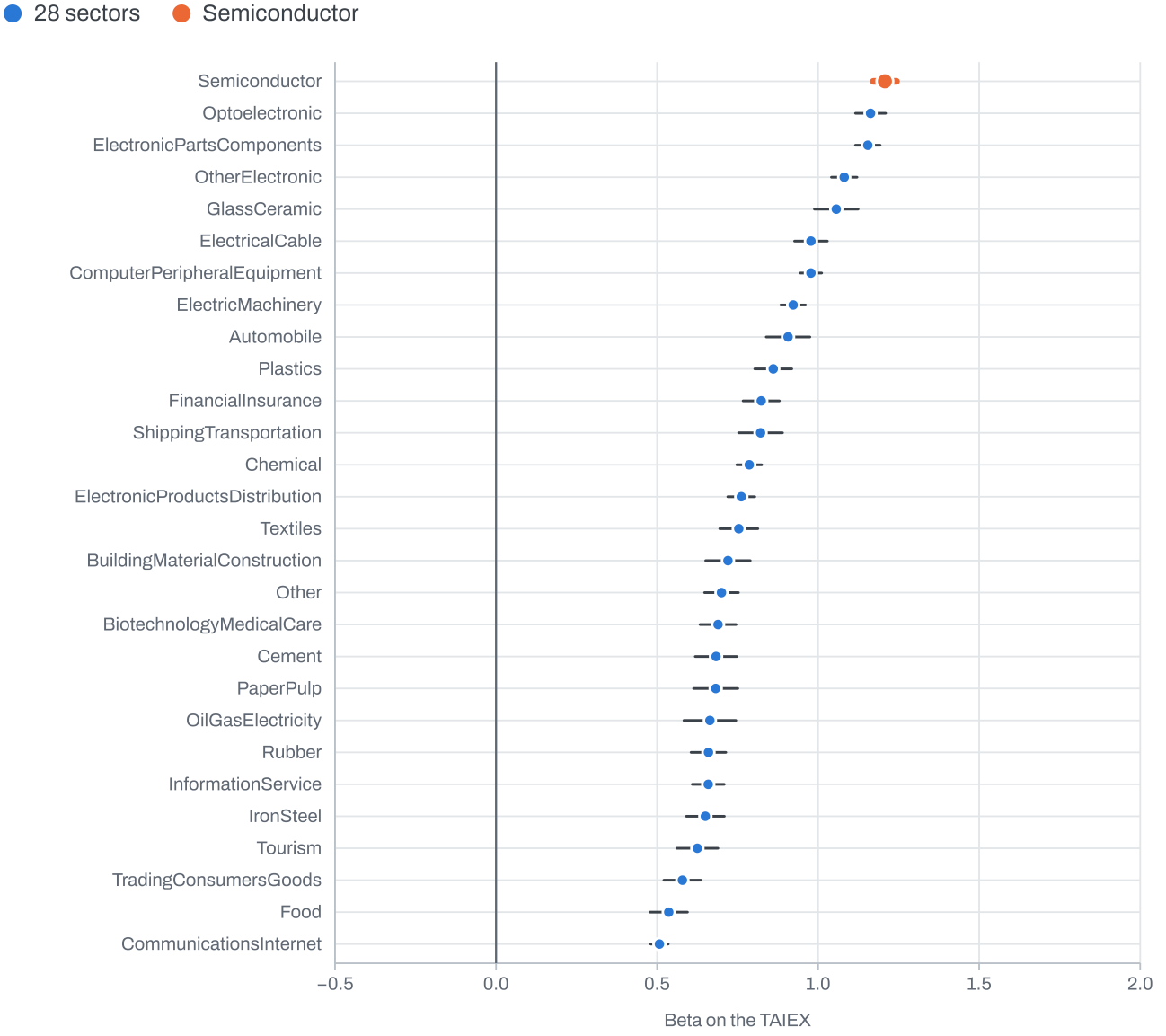


Note: OLS slope and classical 95% band over each window of 250 daily price returns, dated by its last day. The 4,133 windows, the first ending 30 November 2009, are plotted weekly plus the cited extremes. Dashed: the full-sample betas 0.823 and 1.207. Stability check. Source: as Exhibit 1.

The widget shows the 28 market models for each window and standard error, opening on Exhibit 1's fit.

Betas and alphas by window

INTERACTIVE



Beta · Newey–West SE	95% interval for beta	Intercept: bp a day · % a year
1.207 · 0.0185	**1.171 to 1.244**	**2.10 · 5.16%**
t-stat of the intercept	**R-squared · residual SD (% a day)**	**Returns T · Newey–West lag**
2.09	**0.8112 · 0.662**	**4,382 · 9**

Daily price returns of the 28 U28 sector indices on the TAIEX price index (no dividends), in-sample. Rows keep the order of the full-sample beta, so a sector that moves shows it. Intervals are the estimate ± 1.96 standard errors with the method chosen; Newey–West uses the book's lag for the window's size. Every number was computed in advance by data/lm23_charts.py.

EXAMPLE 6

Hedging a sector with TX

A fund holds NT$100 million of the FinancialInsurance index (hypothetical) and shorts TX futures, each contract worth $200F_t$ NT$ (LM8). Find the variance-minimizing hedge ratio, the

contracts on 8 October 2026, and how a first-half ratio performs in the second half.

SOLUTION

Treat TX as the TAIEX, ignoring the basis (LM25). By (16.3) the hedged return has the smallest variance when the hedge ratio equals β, as in (18.6). The minimum is $1-\rho^2$ times the unhedged variance: the hedge removes the share R-squared.

The full-sample beta is 0.82338 (R-squared 0.5916); the last 250 returns give 0.30847 (0.1989). TX settled at $F_t = 49{,}357$ on 8 October 2026 (the 202610 contract), so a contract was worth NT$9,871,400. The notional hedge is 10.13 contracts; the full-sample beta hedges 8.34, the last-250 beta 3.12, about 3.

Out of sample, the first half's beta, 1.1556, cuts the second half's variance by only 0.19%. A ratio of 1 cuts it by 23.09%, and the second half's own 0.5784 by 49.29% (in-sample). Across the 28 sectors, the first-half ratio keeps a median 87% of the second half's in-sample cut. It cuts under 1%, or raises the variance, in five sectors. A regression hedge is only as good as the beta's stability.

EXAMPLE 7

00631L's elasticity, with an error bar

LM2 regressed 00631L's daily total returns on 0050's over 2,902 common-date pairs and found a slope of 1.887. Give its standard errors, test the designed multiple 2, and compare the calendar years 2015–2025.

SOLUTION

The slope is 1.8865, with standard errors 0.0081 classical, 0.0154 White and 0.0211 Newey–West ($\ell = 8$). The designed 2 lies $(2 - 1.8865)/0.0211 = 5.38$ Newey–West standard errors away, outside the interval from 1.845 to 1.928.

The yearly slopes run from 1.717 (2024) to 1.984 (2021), with standard errors 0.0759 and 0.0776 ($\ell = 4$). Four of the 11 lie more than 1.96 of their own standard errors from 1.8865, against about 0.55 expected if the slope were constant. The extremes, 2.5 standard errors of the difference apart, prove less: they were picked from 11 years. The fund's measured exposure is below 2 on average, and it moves between years.

導讀 **beta、alpha 與市場模型**

市場模型（market model）把類股報酬對大盤報酬做迴歸（regression）：斜率（slope）是 beta，截距（intercept）是 alpha。CAPM 用含息的超額報酬（total excess return），那條迴歸的截距才是 Jensen's alpha，模型成立時應為零。本模組用的是價格指數（price index），也沒有扣除無風險利率（risk-free rate），所以截距裡混了利率項與股利殖利率（dividend yield）差，必須依 (23.8) 拆開才能談 alpha。beta 也不是固定常數：28 個類股中有 25 個，前後兩半的 beta 差距超過 1.96 個標準誤（standard error）。以 beta 下降最多（1.156 降到 0.578）的金融保險類股為例，前半段估出的避險比例（hedge ratio）在後半段幾乎沒有降低變異數（variance）。避險後剩下的，就是殘差（residual）的風險。

KNOWLEDGE CHECK 5

CommunicationsInternet has $\beta = 0.507$ and a raw intercept of −0.18 bp a day (classical standard error 0.99 bp). By (23.8), how much does a 2% risk-free rate change Jensen's alpha, dividends aside?

ANSWER

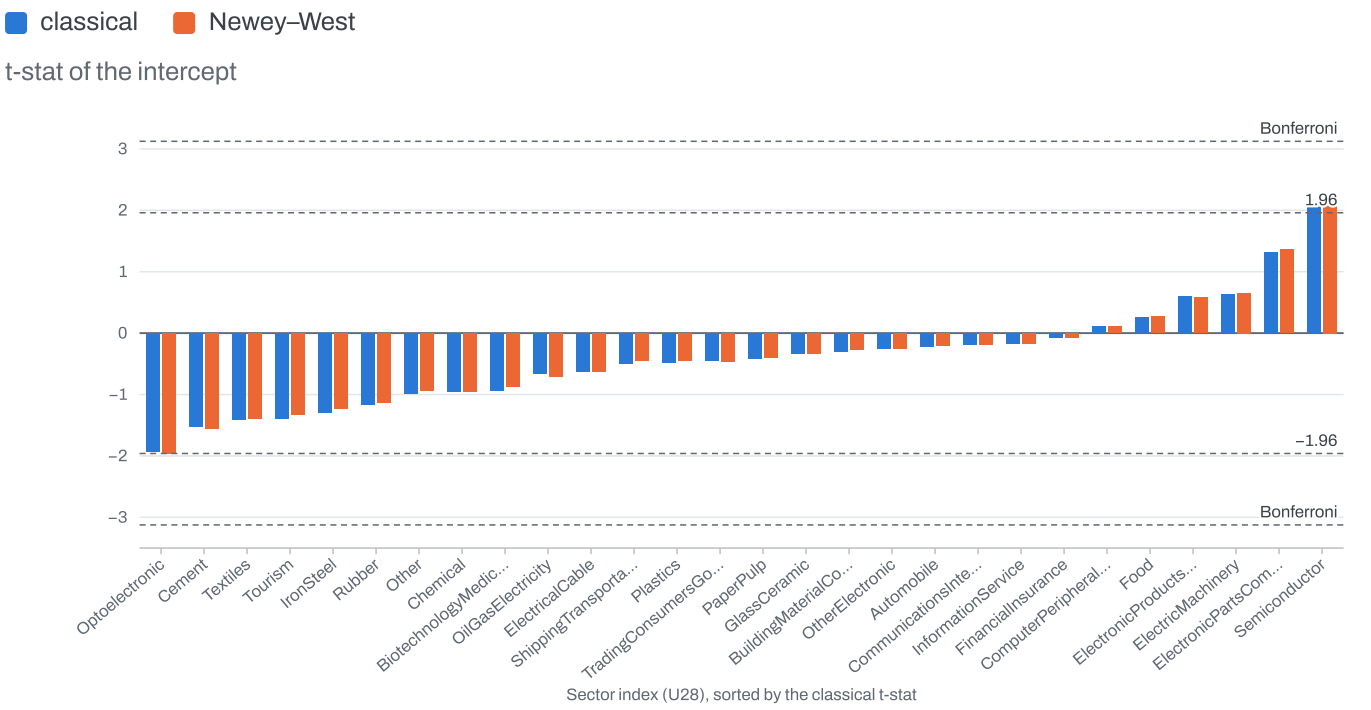
The term $(1-\beta)r_f/A = 0.493 \times 2\%/245.4794$ is 0.40 bp a day, 0.99% a year. Jensen's alpha is the raw intercept minus it, about −0.6 bp a day. A low-beta sector's raw intercept overstates its alpha, here by less than half a standard error.

7 TWENTY-EIGHT ALPHAS

- [ ] test 28 sector alphas with Bonferroni, Holm and Benjamini–Hochberg, and explain how correlation across sectors changes the count of false positives

Exhibit 7 shows the 28 intercept t-stats, classical and Newey–West. At the 5% level one is significant with the classical and White errors, Semiconductor's. Newey–West's errors add Optoelectronic, at −1.965. Chance alone predicts $28 \times 0.05 = 1.4$, whatever the dependence, because an expected count sums the tests' probabilities.

Exhibit 7: The 28 intercept t-stats, classical and Newey–West



Note: market models of U28's daily price returns, sorted by the classical t-stat; Newey–West with $\ell = 9$. Lines at ±1.96 and at Bonferroni's ±3.12 for 28 two-sided tests at 5%. In-sample; one family of 28 tests. Source: as Exhibit 1.

LM21 gives three corrections for a family of tests. Bonferroni tests each at $5\%/28$, a p-value below 0.00179 or $|\text{t-stat}| > 3.12$, (21.6). Holm's steps (21.7) and Benjamini–Hochberg's false discovery rate (21.8), both at 5%, are less strict. The smallest p-value, Semiconductor's 0.036, fails all three, with every standard error.

The sectors are not independent tests. Their residuals correlate from −0.430 to 0.462 across the 378 pairs, with mean 0.112. All 28 fits share one regressor, so under (23.2)'s conditions their intercepts' errors inherit these correlations.

EXAMPLE 8

How many false alphas?

Under the null that all 28 alphas are zero, simulate the count of $|\text{t-stat}| > 1.96$ with the residual correlation matrix C, and compare independent tests.

SOLUTION

Experiment 1 draws 100,000 vectors of 28 null t-stats from $\mathcal{N}_{28}(\mathbf{0}, C)$, colouring standard normals by C's Cholesky factor as in (16.7). It uses `default_rng([1, 20261011])` and U28's column order (Problem 12 gives the call). The count has mean 1.41 (Monte Carlo SE 0.005), as for independent tests. Its standard deviation is 1.46 (SE 0.006), against $\sqrt{28 \times 0.05 \times 0.95} = 1.153$.

At least one test passes with probability 0.696 (SE 0.0015), against $1 - 0.95^{28} = 0.7622$ for independent tests, and three or more with 0.178 (SE 0.0012), against 0.163. Correlation keeps the mean count but makes passes cluster.

The case's three questions now have answers. Semiconductor's beta is measured to about ±0.04 (Newey–West, 95%), but it drifts. Its intercept, 1.98 bp after interest and equal dividend yields, is one draw from 28, and no correction rejects. Searching many sectors, rules or windows and keeping the best is the data snooping that LM34 treats.

導讀 **28 個 alpha 與多重假設檢定**

同一份資料在 5% 顯著水準（significance level）下做 28 次檢定（test），即使真實的 alpha 全是零，平均也有 1.4 個 t 統計量（t-statistic）的絕對值超過 1.96，所以看到一個顯著並不稀奇。多重假設檢定（multiple testing）控制族系誤差率（family-wise error rate）或偽發現率（false discovery rate）：Bonferroni 校正（Bonferroni correction）把每個檢定的門檻（threshold）降到 0.05/28，Holm 逐步放寬，Benjamini–Hochberg 控制偽發現率。三種方法都沒有拒絕任何類股。類股殘差（residual）彼此相關（correlated），不改變錯誤拒絕（false rejection）的平均個數，卻讓「至少一個」的機率（probability）從 0.762 降到 0.696。把許多類股、規則或期間都試一遍再挑最好的，就是資料偵伺（data snooping）。

KNOWLEDGE CHECK 6

What p-value and what |t-stat| does Bonferroni require of each of 28 two-sided tests at a family-wise level of 5%? Does the expected count 1.4 need independent tests?

ANSWER

Each test runs at $0.05/28 = 0.00179$, so |t-stat| must exceed $\Phi^{-1}(1 - 0.025/28) = 3.12$. The expected count needs no independence: it sums 28 expectations of 0.05 each. Its spread does depend on independence, as Example 8 shows.

8 A FACTOR NAMED IN ADVANCE

- [ ] estimate a factor model with a factor named in advance, and diagnose the collinearity it brings

LM11 found principal components and warned that nothing makes its second component "electronics". An economic factor model names its factors before looking. Here the factor is the Electronic index's return minus the TAIEX's, constructed by this module; Electronic aggregates the electronics sectors, Semiconductor among them.

$$\begin{aligned} R_{i,t} = \alpha &+ \beta_{\text{mkt}} R_{\text{mkt},t} \\ &+ \beta_{\text{el}} \left(R_{\text{el},t} - R_{\text{mkt},t} \right) + \varepsilon_t \end{aligned}$$

Why not regress on the TAIEX and Electronic directly? Their returns correlate 0.9696. With two regressors of sample correlation $\hat{\rho}$, each slope's variance is its value with uncorrelated regressors times the **variance inflation factor**

$$\text{VIF} = \frac{1}{1 - \hat{\rho}^2}. \tag{23.9}$$

The factor comes from (23.2): the slopes' block of $(X^\top X)^{-1}$ is $(T\hat{\Sigma})^{-1}$ by (12.3), and inverting a 2×2 correlation matrix gives diagonal entries $1/(1-\hat{\rho}^2)$. For the raw pair the VIF is 16.7, and the condition number of their correlation matrix, $(1+\hat{\rho})/(1-\hat{\rho})$, is 65; (12.6) relates it to the data's. The factor correlates only 0.365 with the TAIEX: VIF 1.15, condition number 2.15.

The raw pair and the factor pair give the same fitted values and residuals (Problem 10).

EXAMPLE 9

Two sectors on two factors

Fit the two-factor model to Semiconductor and FinancialInsurance, compare R-squared with the market model's, and refit Semiconductor on the raw pair (TAIEX, Electronic).

SOLUTION

Semiconductor's coefficients are 1.037 on the market and +1.552 on the factor, and R-squared rises from 0.8112 to 0.9163 (Exhibit 8). FinancialInsurance loads −1.580 on the factor, and its R-squared rises from 0.5916 to 0.7624: with the market held fixed, finance moves against electronics.

On the raw pair, Semiconductor gets −0.515 on the TAIEX and +1.552 on Electronic, with the same residuals, and $-0.515 + 1.552 = 1.037$. The negative market coefficient is collinearity, not an economic finding. Its classical standard error is 0.024, against 0.006 in the factor pair.

Part of the loading is mechanical: Electronic contains Semiconductor, so the fit partly regresses Semiconductor on itself. Across sectors, 27 of 28 loadings have $|\text{t-stat}| > 1.96$ by either standard error; the weakest, CommunicationsInternet's, has −1.05 classical and −0.68 Newey–West. The halves' 28 loadings have cosine 0.807, against 0.297 for LM11's second component across the same halves.

Exhibit 8: One factor against two: six sectors

Sector	Beta, market model	R-squared	β_{mkt} (SE)	β_{el} (SE)	R-squared, two factors
ElectronicPartsComponents	1.154	0.6842	1.105 (0.019)	+0.452 (0.058)	0.6924
ComputerPeripheralEquipment	0.978	0.6561	0.950 (0.017)	+0.249 (0.057)	0.6594
Semiconductor	1.207	0.8112	1.037 (0.009)	+1.552 (0.033)	0.9163
CommunicationsInternet	0.507	0.4368	0.511 (0.013)	−0.033 (0.048)	0.4370
FinancialInsurance	0.823	0.5916	0.997 (0.022)	−1.580 (0.053)	0.7624
Plastics	0.861	0.3797	1.048 (0.030)	−1.706 (0.078)	0.4966

Note: daily price returns of U28; the electronics factor $R_{\text{el},t} - R_{\text{mkt},t}$ is constructed, and the four electronics sectors are inside Electronic. Standard errors are Newey–West ($\ell = 9$); in-sample. Source: as Exhibit 1, with Electronic from the same dataset.

Cross-sectional asset-pricing tests (Fama and MacBeth 1973), a joint test of the 28 alphas, betas shrunk toward 1 and lagged market terms are outside this book. Factor risk in a portfolio is LM27's bookkeeping and LM30's estimation problem.

KNOWLEDGE CHECK 7

TAIEX and Electronic returns correlate 0.9696. By what factor does collinearity multiply each slope's variance? What if the correlation is 0.365?

ANSWER

By (23.9), $1/(1-0.9696^2) = 16.7$: each standard error is about four times its value with uncorrelated regressors. With 0.365 the factor is $1/(1-0.365^2) = 1.15$.

9 REGRESSION IN PYTHON

- [ ] compute OLS with classical and robust standard errors in NumPy, and check them against statsmodels with every option passed explicitly

NumPy computes every number here. The checks use statsmodels with every option passed, because its defaults differ from the book's.

PYTHON

```python
import numpy as np
import statsmodels.api as sm
from statsmodels.stats.multitest import multipletests

# y: a sector's returns, x: the TAIEX's (U28, T = 4,382)
X = np.column_stack([np.ones(len(x)), x])
b = np.linalg.lstsq(X, y, rcond=None)[0]
e = y - X @ b
s2 = e @ e / (len(y) - X.shape[1])          # divisor T - d
XtXi = np.linalg.inv(X.T @ X)
se_classical = np.sqrt(s2 * np.diag(XtXi))
u = X * e[:, None]
se_white = np.sqrt(np.diag(XtXi @ (u.T @ u) @ XtXi))   # HC0

white = sm.OLS(y, X).fit(cov_type="HC0")
nw = sm.OLS(y, X).fit(cov_type="HAC",
                      cov_kwds={"maxlags": 9, "use_correction": False})
print(nw.bse, nw.use_t)                     # use_t False: normal p-values
# pvals: the 28 intercepts' two-sided p-values (Problem 12)
holm = multipletests(pvals, alpha=0.05, method="holm")[0]
```

`get_robustcov_results()` without arguments returns HC1, which multiplies HC0's variance by $T/(T-d)$. In `OLS.fit` the HAC option needs `maxlags` and leaves `use_correction` False; `cov_hac` defaults to True. `multipletests` defaults to Holm–Šidák (`"hs"`), so name the method.

IN PYTHON **Companion lab**

The Polars-first notebook `LM23_lab.ipynb` rebuilds the module's exhibits and computational examples and solves the computational practice problems. It ends with a walk-forward hedge: one rule, fixed in advance and judged out of sample.

The notebook reads FinMind files that you download once with your own token. From the repository root, run `uv run python data/fetch_finmind.py`, then `uv run python data/make_lab_data.py`. FinMind's licence does not allow the book to redistribute the data. After that, the notebook runs offline.

KNOWLEDGE CHECK 8

On FinancialInsurance's market model, `sm.OLS(y, X).fit(cov_type="HAC", cov_kwds={"maxlags": 9, "use_correction": True})` gives a beta standard error of 0.02885, against the book's 0.0288448. Where does the difference come from?

ANSWER

The correction multiplies the covariance by $T/(T-d)$, so the standard error grows by $\sqrt{4382/4380} = 1.000228$: $0.0288448 \times 1.000228 = 0.02885$. The book uses no correction, the default of `OLS.fit`.

SUMMARY

- OLS is unbiased when the errors have mean zero given X, (23.2); s_ε^2 divides RSS by $T-d$, (23.3).
- Classical standard errors (23.4) assume homoskedastic, uncorrelated errors; hat values measure influence, (23.5).
- White's sandwich (23.6) allows heteroskedasticity, and Newey–West (23.7) autocorrelation too.
- A Newey–West standard error that grows with the lag signals a drifting coefficient.
- A raw intercept on price indices is not Jensen's alpha, (23.8), and betas move.
- One significant intercept among 28 is what chance gives.
- A factor named in advance avoids collinearity, measured by (23.9).

PRACTICE PROBLEMS

1. A regression of $T = 12$ monthly returns on a constant and one regressor has $\text{RSS} = 0.0240$. Its s_ε is *closest* to:

   A. 4.47%

   B. 4.90%

C. 4.67%

2. A simple regression has $s_\varepsilon = 1\%$, $s_x = 1.2\%$ (divisor T) and $T = 1{,}000$. The classical $\mathrm{SE}(\hat\beta)$ is *closest* to:

A. 0.0264

B. 0.833

C. 0.000833

3. Semiconductor's market model has $\hat\beta = 1.2073$, with standard errors 0.0088 classical and 0.0185 Newey–West ($\ell = 9$). The Newey–West t-stat of $H_0 : \beta = 1$ is *closest* to:

A. 65.3

B. 23.6

C. 11.2

4. Daily errors and regressor returns are iid and mutually independent, and a regression uses their overlapping 20-day sums. The slope's true standard error is *closest* to how many times the classical one?

A. 3.7

B. 1

C. 4.5

5. A price-index market model gives $\beta = 0.6$ and an intercept of 1.0 bp a day. Take $r_f = 2\%$ a year and $A = 245.48$, and ignore dividends ($q_i = q_{\mathrm{mkt}} = 0$). By (23.8), Jensen's alpha is *closest* to:

A. 1.326 bp

B. 0.185 bp

C. 0.674 bp

6. Five tests give the p-values 0.004, 0.011, 0.019, 0.030 and 0.20. At a family-wise level of 5%, Holm's procedure (21.7) rejects how many?

A. 1

B. 2

C. 4

7. Derive (23.2) from (16.4). Then show $\mathbb{E}[\mathrm{RSS} \mid X] = (T-d)\sigma_\varepsilon^2$ from $\mathrm{tr}(I - \Pi) = T - d$.

8. In a simple regression, show that $h_t = \frac{1}{T} + (x_t - \bar x)^2/(T s_x^2)$ and that the h_t sum to 2. Compute 7 April 2025's hat value in FinancialInsurance's 2025 fit from the TAIEX's mean 0.1062%, $s_x = 1.5405\%$ (divisor 243) and $x_t = -9.6997\%$, to four decimals. *(Python)* Recompute all 243 hat values.

9. A fund holds NT$50 million in a sector with $\beta = 0.75$ and R-squared 0.5 on the TAIEX (hypothetical), and TX settles at $F_t = 20{,}000$, NT$200 a point. Find the variance-minimizing number of contracts, the notional hedge, and the hedged standard deviation as a share of the unhedged one. Why does an in-sample R-squared overstate the reduction?

10. Show that regressing on $(R_{\text{mkt}}, R_{\text{el}})$ and on $(R_{\text{mkt}}, R_{\text{el}} - R_{\text{mkt}})$ gives the same fitted values and the same electronics coefficient. Show that the factor pair's market coefficient is the sum of the raw pair's slopes. Check it on Semiconductor, and give each pair's VIF from the correlations 0.9696 and 0.365.

11. *(Python)* Fit FinancialInsurance's market model on U28 in NumPy and in statsmodels with `cov_type` `"nonrobust"`, `"HC0"` and `"HAC"` (`cov_kwds={"maxlags": 9, "use_correction": False}`). Give $\hat{\beta}$ to four decimals, the three standard errors and the HAC value with `use_correction=True`. Why does `get_robustcov_results()` give another number?

12. *(Python)* Compute the 28 classical intercept t-stats, their two-sided normal p-values and `multipletests` with `method="bonferroni"`, `"holm"` and `"fdr_bh"`. Then run experiment 1 for $\Pr(\text{at least one } |\text{t-stat}| > 1.96)$ and its Monte Carlo standard error. Use `default_rng([1, 20261011])` and 100,000 draws as `rng.standard_normal((100_000, 28)) @ L.T` with `L = np.linalg.cholesky(C)` and the sectors in U28's column order, Python's `sorted` of their FinMind names (ElectricMachinery before ElectricalCable).

SOLUTIONS

1. **B** is correct: $s_\varepsilon = \sqrt{0.0240/(12-2)} = 4.90\%$ by (23.3). A divides by T, which ignores the two fitted coefficients. C divides by $T-1$, LM20's s_{u}, which allows for one.

2. **A** is correct: $0.01/(0.012\sqrt{1000}) = 0.0264$ by (23.2). B omits $\sqrt{T}$. C divides by T, not $\sqrt{T}$.

3. **C** is correct: $(1.2073 - 1)/0.0185 = 11.2$. A tests $\beta = 0$, $1.2073/0.0185 = 65.3$. B uses the classical error, $0.2073/0.0088 = 23.6$.

4. **A** is correct: by the Deep Dive of Section 5 the factor is

$$\sqrt{\frac{2m^2+1}{3m}} = \sqrt{\frac{801}{60}} = 3.65.$$

   B ignores the overlap. C, $\sqrt{20} = 4.47$, is the non-overlapping sums' standard error relative to the overlapping classical one; overlap keeps more information, so the true factor is smaller.

5. **C** is correct. With $q_i = q_{\text{mkt}} = 0$, (23.8) gives

$$\begin{aligned} \alpha_{\text{J}} &= \alpha - (1-\beta) r_f / A \\ &= 1.0 - 0.4 \times 0.02/245.48 \times 10^4 \\ &= 1.0 - 0.326 = 0.674 \text{ bp}. \end{aligned}$$

A adds the term instead. B subtracts $r_f/A = 0.815$ bp, ignoring $1-\beta$.

6. **B** is correct. Holm compares the ordered p-values with $0.05/5 = 0.01$, $0.05/4 = 0.0125$ and $0.05/3 = 0.0167$: 0.004 and 0.011 pass, 0.019 fails, and Holm stops. A is Bonferroni's count (only 0.004 is below 0.01). C is Benjamini–Hochberg's at a 5% false discovery rate: 0.030 is below $4 \times 0.05/5 = 0.04$. statsmodels' default, Holm–Šidák, also gives 2 here, but its thresholds differ, so name the method.

7. Substituting $\mathbf{y}$ gives $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (X^\top X)^{-1} X^\top \boldsymbol{\varepsilon}$. By (16.4) and condition 3,

$$\begin{aligned} &\operatorname{Var}(\hat{\boldsymbol{\beta}} \mid X) \\ &\quad = (X^\top X)^{-1} X^\top (\sigma_\varepsilon^2 I) X (X^\top X)^{-1} \\ &\quad = \sigma_\varepsilon^2 (X^\top X)^{-1}, \end{aligned}$$

   which is (23.2). Next, RSS is a scalar, so it equals its trace:

$$\mathrm{RSS} = \boldsymbol{\varepsilon}^\top (I - \Pi) \boldsymbol{\varepsilon} = \operatorname{tr}\left((I - \Pi) \boldsymbol{\varepsilon} \boldsymbol{\varepsilon}^\top \right).$$

   Taking expectations given X, condition 3 gives

$$\begin{aligned} \mathbb{E}[\mathrm{RSS} \mid X] &= \operatorname{tr}\left((I - \Pi) \sigma_\varepsilon^2 I \right) \\ &= (T - d) \sigma_\varepsilon^2. \end{aligned}$$

8. With $\mathbf{x}_t = (1, x_t)^\top$, the matrix $X^\top X$ has entries T, $T\bar{x}$, $T\bar{x}$ and $T(s_x^2 + \bar{x}^2)$, with determinant $T^2 s_x^2$. Hence

$$\begin{aligned} h_t &= \frac{T(s_x^2 + \bar{x}^2) - 2T\bar{x}x_t + T x_t^2}{T^2 s_x^2} \\ &= \frac{1}{T} + \frac{(x_t - \bar{x})^2}{T s_x^2}. \end{aligned}$$

   Summing over t gives $1 + T s_x^2 / (T s_x^2) = 2 = d$. For 7 April 2025,

$$h_t = \frac{1}{243} + \frac{9.8059^2}{243 \times 1.5405^2} = 0.1709,$$

   21 times the average $2/243$. The lab computes all 243 from Π.

9. The contracts are $0.75 \times 50{,}000{,}000 / (200 \times 20{,}000) = 9.375$, about 9, against 12.50 for the notional hedge. The hedged variance is $1 - 0.5$ of the unhedged, so the standard deviation is $\sqrt{0.5} = 0.7071$ of it. In-sample, the fitted beta is the best ratio by construction; a moving beta, as in Example 6, cuts less out of sample.

10. Each design's columns are combinations of the other's, since $R_{\mathrm{el}} = (R_{\mathrm{el}} - R_{\mathrm{mkt}}) + R_{\mathrm{mkt}}$. So both span the same space, and Π, the fitted values and the residuals agree. Equating the fits day by day,

$$\begin{aligned} &\beta_{\mathrm{mkt}}^{\mathrm{raw}} R_{\mathrm{mkt}} + \beta_{\mathrm{el}}^{\mathrm{raw}} R_{\mathrm{el}} \\ &\quad = \beta_{\mathrm{mkt}} R_{\mathrm{mkt}} + \beta_{\mathrm{el}} (R_{\mathrm{el}} - R_{\mathrm{mkt}}), \end{aligned}$$

gives $\beta_{\text{el}} = \beta_{\text{el}}^{\text{raw}}$ and $\beta_{\text{mkt}} = \beta_{\text{mkt}}^{\text{raw}} + \beta_{\text{el}}^{\text{raw}}$. For Semiconductor, $-0.515 + 1.552 = 1.037$. The VIFs are $1/(1 - 0.9696^2) = 16.7$ and $1/(1 - 0.365^2) = 1.15$.

11. Both give $\hat{\beta} = 0.8234$, with standard errors 0.0103 classical, 0.0183 HC0 and 0.0288 HAC; NumPy and statsmodels agree to 12 digits. With `use_correction=True` the HAC value is 0.02885. `get_robustcov_results()` defaults to HC1, which scales HC0's variance by $T/(T-2)$.

12. The t-stats run from −1.935 (Optoelectronic) to 2.097 (Semiconductor); the smallest p-value is 0.036. Bonferroni, Holm and Benjamini–Hochberg reject none. Holm's smallest adjusted p-value is 1.000 and Benjamini–Hochberg's 0.742. Experiment 1 gives $\Pr(\text{at least one}) = 0.696$ with Monte Carlo SE 0.0015, against 0.7622 for independent tests.

GLOSSARY

Term	中文	Meaning
Regression	迴歸	A model $\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ of a response on regressors, (23.1).
Ordinary least squares (OLS)	普通最小平方法	The estimate $(X^\top X)^{-1} X^\top \mathbf{y}$, which minimizes RSS.
Residual	殘差	$\hat{\boldsymbol{\varepsilon}} = \mathbf{y} - X\hat{\boldsymbol{\beta}}$.
Unbiased estimator	不偏估計量	An estimator whose expectation is the parameter, (23.2), (23.3).
Gauss–Markov theorem	高斯-馬可夫定理	Under (23.2)'s conditions, OLS has the smallest variance among linear unbiased estimators.
Best linear unbiased estimator	最佳線性不偏估計量	OLS under the Gauss–Markov conditions.
Standard error	標準誤	The estimated standard deviation of an estimator.
Confidence interval	信賴區間	$\hat{\beta}_j \pm 1.96\,\mathrm{SE}(\hat{\beta}_j)$, (23.4).
R-squared	判定係數	The share of the centred response's squared length that the fit reproduces, (12.4).
Hat value	—	h_t, the diagonal entry of Π in row t: a point's influence on its own fit, (23.5).
Heteroskedasticity	變異數非齊一性	Error variances that differ across observations.
Robust standard error	穩健標準誤	A standard error from the sandwich (23.6), such as White's (HC0).
HAC estimator	變異數非齊一與自我相關一致估計量	Newey–West, (23.7): robust to heteroskedasticity and autocorrelation.
Autocorrelation	自我相關	Correlation of a series with its own past.

Term	中文	Meaning
Long-run variance	—	The limit of the filling of (23.6) over T for autocorrelated rows, which (23.7) estimates.
Market model	市場模型	$R_{i,t} = \alpha + \beta R_{\mathrm{mkt},t} + \varepsilon_t$.
Jensen's alpha	—	The market model's intercept on total excess returns; (23.8) links it to a raw intercept.
Multiple testing	多重假設檢定	Running many tests on the same data.
Family-wise error rate	族系誤差率	The probability of at least one false rejection.
False discovery rate	偽發現率	The expected share of false rejections among all rejections.
Bonferroni correction	Bonferroni 校正	Each test of a family at the family's level divided by the number of tests, (21.6).
Data snooping	資料偵伺	Searching many models or samples and keeping the best.
Collinearity	共線性	Highly correlated regressors; it inflates the slopes' variances by (23.9).

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